

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	$16 + (21-1) \times d = 24 \Rightarrow d = \dots$	M1	1.1b
	$d = 0.4$	A1	1.1b
	Answer only scores both marks.		
	(2)		
(b)	$S_n = \frac{1}{2}n\{2a + (n-1)d\} \Rightarrow S_{500} = \frac{1}{2} \times 500\{2 \times 16 + 499 \times "0.4"\}$	M1	1.1b
	$= 57900$	A1	1.1b
	Answer only scores both marks		
	(2)		
(b) Alternative using $S_n = \frac{1}{2}n\{a + l\}$			
	$l = 16 + (500-1) \times "0.4" = 215.6 \Rightarrow S_{500} = \frac{1}{2} \times 500\{16 + "215.6"\}$	M1	1.1b
	$= 57900$	A1	1.1b
(4 marks)			

Notes
(a) M1: Correct strategy to find the common difference – must be a correct method using $a = 16$, and $n = 21$ and the 24. The method may be implied by their working. If the AP term formula is quoted it must be correct, so use of e.g. $u_n = a + nd$ scores M0 A1: Correct value. Accept equivalents e.g. $\frac{8}{20}, \frac{4}{10}, \frac{2}{5}$ etc. (b) M1: Attempts to use a correct sum formula with $a = 16$, $n = 500$ and their numerical d from part (a) If a formula is quoted it must be correct (it is in the formula book) A1: Correct value Alternative: M1: Correct method for the 500th term and then uses $S_n = \frac{1}{2}n\{a + l\}$ with their l A1: Correct value Note that some candidates are showing implied use of $u_n = a + nd$ by showing the following: (a) $d = \frac{24-16}{21} = \frac{8}{21}$ (b) $S_{500} = \frac{1}{2} \times 500 \left\{ 2 \times 16 + 499 \times \frac{8}{21} \right\} = 55523.80952\dots$ This scores (a) M0A0 (b) M1A0

Q2.

Question	Scheme	Marks	AOs
(i)	States that $S = a + (a + d) + \dots + (a + (n-1)d)$	B1	1.1a
	$S = a + \dots + (a + d) + \dots + (a + (n-1)d)$ $S = (a + (n-1)d) + (a + (n-2)d) + \dots + a$	M1	3.1a
	Reaches $2S = n \times (2a + (n-1)d)$ And so proves that $S = \frac{n}{2}[2a + (n-1)d]$ *	A1*	2.1
		(3)	
(ii)	(a) $S = 10 + 9.20 + 8.40 + \dots$		
	$64 = \frac{n}{2}(20 - 0.8(n-1))$ o.e	M1	3.1b
	$128 = 20n - 0.8n^2 + 0.8n$ $0.8n^2 - 20.8n + 128 = 0$ $n^2 - 26n + 160 = 0$ *	A1*	2.1
		(2)	
	(b) $n = 10, 16$	B1	1.1b
		(1)	
	(c) 10 weeks with a minimal correct reason. E.g. • He has saved up the amount by 10 weeks so he would not save for another 6 weeks • You would choose the smaller number • He starts saving negative amounts (in week 14) so 16 does not make sense	B1	2.3
		(1)	
(7 marks)			
Notes:			

(i)

B1: Correctly writes down an expression for the key terms S or S_n including $S =$ or $S_n =$

Allow a minimum of 3 correct terms including the first and last terms, and no incorrect terms.

Score for S or $S_n = a + (a + d) + \dots + (a + (n-1)d)$ with + signs, not commas

If the series contains extra terms that should not be there E.g

$S = a + (a + d) + \dots + (a + nd) + (a + (n-1)d)$ score B0

M1: For the key step in reversing the terms and adding the two series.

Look for a minimum of two terms, including a and $a + (n-1)d$, the series reversed with evidence of adding, for example $2S =$ Condone the extra incorrect terms (see above) appearing.

Can be scored when terms are separated by commas

A1*: Shows correct work (no errors) with all steps shown leading to given answer.

There should be no incorrect terms. A minimum of 3 terms should be shown in each sum

The solution below is a variation of this.

$$S = a + (a+d) + \dots + l$$

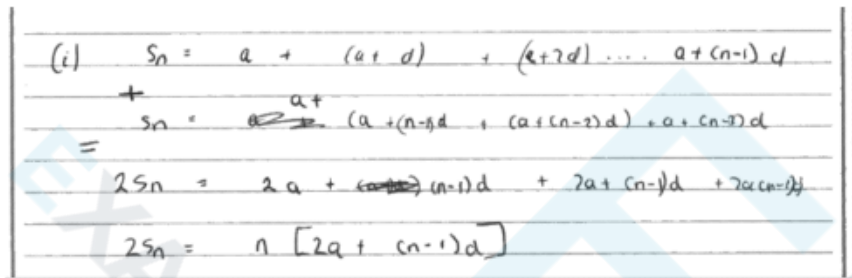
$$S = l + (l-d) + \dots + a$$

$$2S = n(a+l)$$

$$S = \frac{n}{2}(a+l) = \frac{n}{2}(a + a + (n-1)d) = \frac{n}{2}(2a + (n-1)d)$$

B1 and A1 are not scored until the last line, M scored on line 3

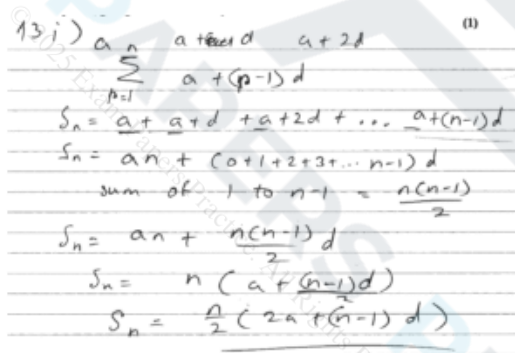
The following scores B1 M0 A0 as the terms in the second sum are not reversed



(i) $S_n = a + (a+d) + (a+2d) \dots a + (n-1)d$
 $+ S_n = a + (a+(n-1)d) + (a+(n-2)d) + \dots + a + (n-1)d$
 $= 2S_n = 2a + (n-1)d + 2a + (n-1)d + \dots + 2a + (n-1)d$
 $2S_n = n[2a + (n-1)d]$

SC in (a) Scores B1 M0 A0.

They use $0+1+2+\dots+(n-1) = \frac{n(n-1)}{2}$ which relies on the quoted proof.



13 i) $\sum_{p=1}^n a + (p-1)d$ (ii)
 $S_n = a + a + d + a + 2d + \dots + a + (n-1)d$
 $S_n = an + (0+1+2+\dots+n-1)d$
 sum of 1 to $n-1 = \frac{n(n-1)}{2}$
 $S_n = an + \frac{n(n-1)}{2}d$
 $S_n = \frac{n}{2}(2a + (n-1)d)$
 $S_n = \frac{n}{2}(2a + (n-1)d)$

(ii) (a)

M1: Uses the information given to set up a correct equation in n .

The values of S , a and d need to be correct and used within a correct formula

Possible ways to score this include unsimplified versions $64 = \frac{n}{2}(2 \times 10 + (n-1) \times -0.8)$,

$64 = \frac{n}{2}(10 + 10 + (n-1) \times -0.8)$ or versions using pence rather than £'s $6400 = \frac{n}{2}(2000 + (n-1) \times -80)$

Allow recovery for both marks following $64 = \frac{n}{2}(2 \times 10 + (n-1) \times -0.8)$ with an invisible $\times$

A1*: Proceeds without error to the given answer. (Do not penalise a missing final trailing bracket)

Look for at least a line with the brackets correctly removed as well as a line with the terms in n correctly combined

E.g. $64 = \frac{n}{2}(20 + (n-1) \times -0.8) \Rightarrow 64 = 10n - 0.4n^2 + 0.4n \Rightarrow 0.4n^2 - 10.4n + 64 = 0 \Rightarrow n^2 - 26n + 160 = 0$

(ii)(b)

B1: $n = 10, 16$

(ii)(c)

B1: Chooses 10 (weeks) and gives a minimal acceptable reason. The reason must focus on why the answer is 10 (weeks) rather than 16(weeks) or alternatively why it would not be 16 weeks.

Q3.

Question	Scheme	Marks	AOs
(a)	Uses the sequence formula $a_{n+1} = \frac{k(a_n + 2)}{a_n}$ once with $a_1 = 2$	M1	1.1b
	$(a_1 = 2), a_2 = 2k, a_3 = k + 1, a_4 = \frac{k(k+3)}{k+1}$ Finds four consecutive terms and sets a_4 equal to a_1 (oe)	M1	3.1a
	$\frac{k(k+3)}{k+1} = 2 \Rightarrow k^2 + 3k = 2k + 2 \Rightarrow k^2 + k - 2 = 0$ *	A1*	2.1
		(3)	
(b)	States that when $k = 1$, all terms are the same and concludes that the sequence does not have a period of order 3	B1	2.3
		(1)	
(c)	Deduces the repeating terms are $a_{1/4} = 2, a_{2/5} = -4, a_{3/6} = -1,$	B1	2.2a
	$\sum_{n=1}^{80} a_k = 26 \times (2 + -4 + -1) + 2 + -4$	M1	3.1a
	$= -80$	A1	1.1b
		(3)	
(7 marks)			
Notes:			

(a)

M1: Applies the sequence formula $a_{n+1} = \frac{k(a_n + 2)}{a_n}$ seen once.

This is usually scored in attempting to find the second term. E.g. for $a_2 = 2k$ or $a_{1+1} = \frac{k(2+2)}{2}$

M1: Attempts to find $a_1 \rightarrow a_4$ and sets $a_1 = a_4$. Condone slips.

Other methods are available. E.g. Set $a_4 = 2$, work backwards to find a_3 and equate to $k+1$

There is no requirement to see either a_1 or any of the labels. Look for the correct terms in the correct order.

There is no requirement for the terms to be simplified

FYI $a_1 = 2, a_2 = 2k, a_3 = k+1, a_4 = \frac{k(k+3)}{k+1}$ and so $2 = \frac{k(k+3)}{k+1}$

A1*: Proceeds to the given answer with accurate work showing all necessary lines. See MS for minimum (b)

B1: States that when $k=1$, all terms are the same and concludes that the sequence does not have a period of order 3.

Do not accept "the terms just repeat" or "it would mean all the terms of the sequence are 2"

There must be some reference to the fact that it does not have order 3. Accept it has order 1.

It is acceptable to state $a_2 = a_1 = 2$ and state that the sequence does not have order 3

(c)

B1: Deduces the repeating terms are $a_{1/4} = 2, a_{2/5} = -4, a_{3/6} = -1$.

M1: Uses a clear strategy to find the sum to 80 terms. This will usually be found using multiples of the first three terms.

For example you may see $\sum_{r=1}^{80} a_r = \left(\sum_{r=1}^{78} a_r \right) + a_{79} + a_{80} = 26 \times (2 + -4 + -1) + 2 + -4$

$$\text{or } \sum_{r=1}^{80} a_r = \left(\sum_{r=1}^{81} a_r \right) - a_{81} = 27 \times (2 + -4 + -1) - (-1)$$

For candidates who find in terms of k award for $27 \times 2 + 27 \times (2k) + 26 \times (k+1)$ or $80k + 80$

If candidates proceed and substitute $k = -2$ into $80k + 80$ to get -80 then all 3 marks are scored.

A1: -80

Note: Be aware that we have seen candidates who find the first three terms correctly, but then find

$$26 \frac{2}{3} \times (2 + -4 + -1) = 26 \frac{2}{3} \times -3 \text{ which gives the correct answer}$$

but it is an incorrect method and should be scored B1 M0 A0

Q4.

Question	Scheme	Marks	AOs
	$\int x^3 \ln x \, dx = \frac{x^4}{4} \ln x - \int \frac{x^4}{4} \times \frac{1}{x} \, dx$	M1	1.1b
	$= \frac{x^4}{4} \ln x - \frac{x^4}{16} (+c)$	M1 A1	1.1b 1.1b
	$\int_1^{e^2} x^3 \ln x \, dx = \left[\frac{x^4}{4} \ln x - \frac{x^4}{16} \right]_1^{e^2} = \left(\frac{e^8}{4} \ln e^2 - \frac{e^8}{16} \right) - \left(-\frac{1^4}{16} \right)$	M1	2.1
	$= \frac{7}{16} e^8 + \frac{1}{16}$	A1	1.1b
		(5)	
(5 marks)			
Notes:			

M1: Integrates by parts the right way round.

Look for $kx^4 \ln x - \int kx^4 \times \frac{1}{x} \, dx$ o.e. with $k > 0$. Condone a missing dx

M1: Uses a correct method to integrate an expression of the form $\int kx^4 \times \frac{1}{x} \, dx \rightarrow c x^4$

A1: $\int x^3 \ln x \, dx = \frac{x^4}{4} \ln x - \frac{x^4}{16} (+c)$ which may be left unsimplified

M1: Attempts to substitute 1 and e^2 into an expression of the form $\pm px^4 \ln x \pm qx^4$, subtracts and uses $\ln e^2 = 2$ (which may be implied).

A1: $\frac{7}{16} e^8 + \frac{1}{16}$ o.e. Allow $0.4375 e^8 + 0.0625$ or uncanceled fractions. NOT ISW: $7e^8 + 1$ is A0

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You may see attempts where substitution has been attempted.

E.g. $u = \ln x \Rightarrow x = e^u$ and $\frac{dx}{du} = e^u$

M1: Attempts to integrate the correct way around condoning slips on the coefficients

$$\int x^3 \ln x \, dx = \int e^{4u} u \, du = \frac{e^{4u}}{4} u - \int \frac{e^{4u}}{4} \, du$$

M1 A1: $\int x^3 \ln x \, dx = \frac{e^{4u}}{4} u - \frac{e^{4u}}{16} (+c)$

M1 A1: Substitutes 0 and 2 into an expression of the form $\pm pue^{4u} \pm qe^{4u}$ and subtracts

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It is possible to use integration by parts "the other way around"

To do this, candidates need to know or use $\int \ln x \, dx = x \ln x - x$

$$\text{FYI } I = \int x^3 \ln x \, dx = x^3 (x \ln x - x) - \int (x \ln x - x) \times 3x^2 \, dx = x^3 (x \ln x - x) - 3I + \frac{3}{4} x^4$$

$$\text{Hence } 4I = x^4 \ln x - \frac{1}{4} x^4 \Rightarrow I = \frac{1}{4} x^4 \ln x - \frac{1}{16} x^4$$

Score M1 for a full attempt at line 1 (condoning bracketing and coefficient slips) followed by M 1 for line 2 where terms in I o.e. to form the answer.

Q5.

Question	Scheme	Marks	AOs
	(i) $\sum_{r=1}^{16} (3+5r+2^r) = 131\,798$; (ii) $u_1, u_2, u_3, \dots, u_{n+1} = \frac{1}{u_n}, u_1 = \frac{2}{3}$		
(i) Way 1	$\left\{ \sum_{r=1}^{16} (3+5r+2^r) \right\} = \sum_{r=1}^{16} (3+5r) + \sum_{r=1}^{16} (2^r)$	M1	3.1a
	$= \frac{16}{2}(2(8)+15(5)) + \frac{2(2^{16}-1)}{2-1}$	M1	1.1b
	$= 728 + 131\,070 = 131\,798^*$	M1	1.1b
		A1*	2.1
	(4)		
(i) Way 2	$\left\{ \sum_{r=1}^{16} (3+5r+2^r) \right\} = \sum_{r=1}^{16} 3 + \sum_{r=1}^{16} (5r) + \sum_{r=1}^{16} (2^r)$	M1	3.1a
	$= (3 \times 16) + \frac{16}{2}(2(5)+15(5)) + \frac{2(2^{16}-1)}{2-1}$	M1	1.1b
	$= 48 + 680 + 131\,070 = 131\,798^*$	M1	1.1b
		A1*	2.1
	(4)		
(i) Way 3	Sum = $10 + 17 + 26 + 39 + 60 + 97 + 166 + 299 + 560 + 1077 + 2106$ $+ 4159 + 8260 + 16457 + 32846 + 65619 = 131\,798^*$	M1	3.1a
		M1	1.1b
		M1	1.1b
		A1*	2.1
	(4)		
(ii)	$\left\{ u_1 = \frac{2}{3}, u_2 = \frac{3}{2}, u_3 = \frac{2}{3}, \dots \right\}$ (can be implied by later working)	M1	1.1b
	$\left\{ \sum_{r=1}^{100} u_r = \right\} 50\left(\frac{2}{3}\right) + 50\left(\frac{3}{2}\right)$ or $50\left(\frac{2}{3} + \frac{3}{2}\right)$	M1	2.2a
	$= \frac{325}{3} \left(\text{or } 108\frac{1}{3} \text{ or } 108.3 \text{ or } \frac{1300}{12} \text{ or } \frac{650}{6} \right)$	A1	1.1b
		(3)	
(7 marks)			

Notes for Question	
(i)	
M1:	Uses a correct methodical strategy to enable the given sum, $\sum_{r=1}^{16} (3+5r+2^r)$ to be found Allow M1 for any of the following: - expressing the given sum as either $\sum_{r=1}^{16} (3+5r) + \sum_{r=1}^{16} (2^r)$, $\sum_{r=1}^{16} 3 + \sum_{r=1}^{16} (5r) + \sum_{r=1}^{16} (2^r)$ or $\sum_{r=1}^{16} 3 + 5 \sum_{r=1}^{16} r + \sum_{r=1}^{16} (2^r)$ - attempting to find both $\sum_{r=1}^{16} (3+5r)$ and $\sum_{r=1}^{16} (2^r)$ separately (3×16) and attempting to find both $\sum_{r=1}^{16} (5r)$ and $\sum_{r=1}^{16} (2^r)$ separately
M1:	Way 1: Correct method for finding the sum of an AP with $a=8$, $d=5$, $n=16$ Way 2: (3×16) and a correct method for finding the sum of an AP
M1:	Correct method for finding the sum of a GP with $a=2$, $r=2$, $n=16$
A1*:	For all steps fully shown (with correct formulae used) leading to 131 798
Note:	Way 1: Give 2 nd M1 for writing $\sum_{r=1}^{16} (3+5r)$ as $\frac{16}{2}(8+83)$
Note:	Way 2: Give 2 nd M1 for writing $\sum_{r=1}^{16} 3 + \sum_{r=1}^{16} (5r)$ as $48 + \frac{16}{2}(5+80)$ or $48 + 680$
Note:	Give 3 rd M1 for writing $\sum_{r=1}^{16} (2^r)$ as $\frac{2(1-2^{16})}{1-2}$ or $2(2^{16}-1)$ or $(2^{17}-2)$

(i)	
Way 3	
M1:	At least 6 correct terms and 16 terms shown
M1:	At least 10 correct terms (may not be 16 terms)
M1:	At least 15 correct terms (may not be 16 terms)
A1*:	All 16 terms correct and an indication that the sum is 131 798
(ii)	
M1:	For some indication that the next two terms of this sequence are $\frac{3}{2}, \frac{2}{3}$
M1:	For deducing that the sum can be found by applying $50\left(\frac{2}{3}\right) + 50\left(\frac{3}{2}\right)$ or $50\left(\frac{2}{3} + \frac{3}{2}\right)$, o.e.
A1:	Obtains $\frac{325}{3}$ or $108\frac{1}{3}$ or $108.\dot{3}$ or an exact equivalent
Note:	Allow 1 st M1 for $u_2 = \frac{3}{2}$ (or equivalent) and $u_3 = \frac{2}{3}$ (or equivalent)
Note:	Allow 1 st M1 for the first 3 terms written as $\frac{2}{3}, \frac{3}{2}, \frac{2}{3}, \dots$
Note:	Allow 1 st M1 for the 2 nd and 3 rd terms written as $\frac{3}{2}, \frac{2}{3}, \dots$ in the correct order
Note:	Condone $\frac{2}{3}$ written as 0.66 or awrt 0.67 for the 1 st M1 mark
Note:	Give A0 for 108.3 or 108.333... without reference to $\frac{325}{3}$ or $108\frac{1}{3}$ or $108.\dot{3}$